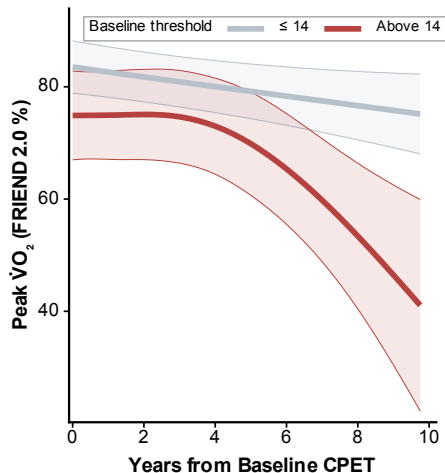
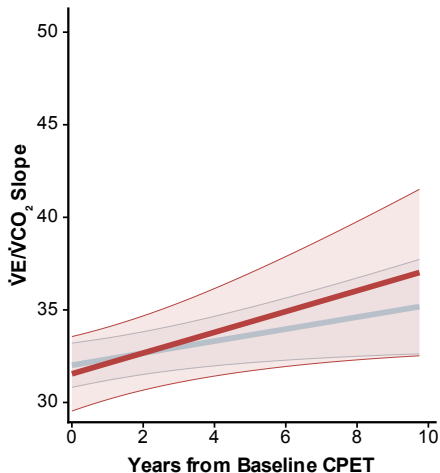


(A) Peak $\dot{V}O_2$ Baseline $E/e' \leq 14$ versus above 14**(B) $\dot{V}E/\dot{V}CO_2$ slope**Baseline $E/e' \leq 14$ versus above 14**(C) Longitudinal Model Summary**

Model component	edf	F statistic	P value
Peak $\dot{V}O_2$			
Time smooth	1.04	5.86	0.016
Threshold group	—	4.10	0.044
Time x threshold group	2.45	3.37	0.018
$\dot{V}E/\dot{V}CO_2$ slope			
Time smooth	1.04	5.61	0.018
Threshold group	—	0.00	1.000
Time x threshold group	1.03	0.77	0.380

Supplemental Figure. Parameter-specific longitudinal cardiopulmonary trajectories stratified by binary baseline E/e' (average) status. **(A)** GAMM-derived Peak $\dot{V}O_2$ trajectories. **(B)** GAMM-derived $\dot{V}E/\dot{V}CO_2$ slope trajectories. **(C)** Summary of the key status and interaction terms for both models. Baseline threshold groups were defined as $E/e' \leq 14$ versus above 14. Longitudinal models were adjusted for age, sex, and BMI.